Computer Organization and Architecture (CSL251) 2025-26 W
LAB-14
Date: 20.04.2026

1. **Write a Verilog code to design a block that routes a single 1-bit input to one of 2^n output lines, based on the value of an n-bit control (select) input.** [20 Marks]

Example, if n=3:

- When sel = 000, the input is routed to output line o0
- When sel = 111, the input is routed to output line o7

All other output lines should remain 0.

Implement the design using:

- If-else statements

Also, write a suitable testbench to verify your design.

2. **Design a Verilog module to generate an even parity bit for a 4-bit binary input.**

- The module takes a 4-bit input number
- It produces a 1-bit output parity
- The output parity should be such that the total number of 1's (including the parity bit) is even

Example:

- If number = 4'b0000 → number of 1's = 0 (even) → parity = 0
- If number = 4'b0001 → number of 1's = 1 (odd) → parity = 1
- If number = 4'b1111 → number of 1's = 4 (even) → parity = 0

Submission requirement:

- Design and testbench code file.
- Simulation diagram.
- Simulation output screenshot

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